

Fig. 7: An actual-size, single-side PCB layout of transmitter circuit for wireless equipment control

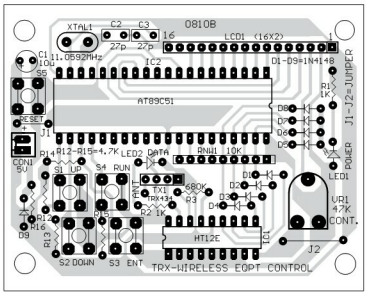


Fig. 8: Component layout for the PCB in Fig. 7

continuously with its local addresses. If no error or unmatched codes are found, the input data codes are decoded and transferred to the output pins. VT pin (valid transmission) goes high to indicate a valid transmission. The HT12D provides four latch-type data pins whose data remains unchanged until new data is received.

Data pins D8 through D11 (pins 10 through 13) of the decoder send 4-bit data to CD4519 multiplexer IC5.

CD4519 multiplexer. This IC provides four multiplexing circuits with common select inputs (S_A and S_B); each circuit contains two inputs (An, Bn) and one output (On). It may be used to select 4-bit information from one of the two sources.

There are eight input lines (A0 through A3 and B0 through B3), of which four (A0 through A3) are permanently connected to Vcc through resistor R19, while the rest four (B0 through B3) are connected to the data output lines of the decoder (IC4).

The select inputs can be connected to either Vcc or VT pin (pin 17) for latch or momentary mode-selection section. Jumper switch (JS) is used to select between latch and momentary operation. When latch mode is selected, data present at the output pins is latched, i.e., they remain the same and the respective relay energises until the next change is made in the mode selection. When momentary mode is selected, data present at the output pins is available as long as VT pin remains active-high. As soon as VT pin becomes active-low, the respective relay de-energises.

The latched output data from multiplexer CD4519 is fed to relay driver IC ULN2003, to control up to four devices through the relays (RL1 through RL4). VT pin is connected to LED4 through IC6 to indicate the status of VT signal when it is active-high.

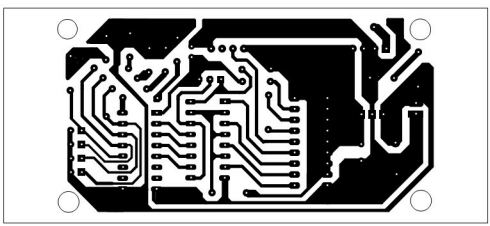


Fig. 9: An actual-size, single-side PCB layout of receiver circuit for wireless equipment control

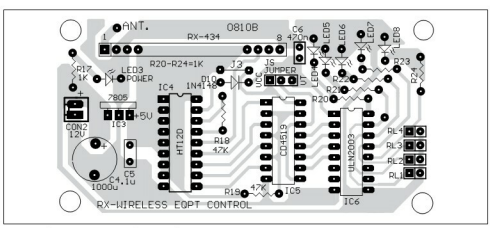


Fig. 10: Component layout for the PCB in Fig. 9